Md Monirul Islam

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Education

- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
  B.Sc. in Electrical and Electronic Engineering *March 2018 – May 2023*
  CGPA: 3.85/4.00 **(3.98/4.00 in the final year)**, ranked **25 of 189**, awarded with Honours
  Relevant Courses: *Electrical & Electronic Circuits, Calculus I & II, Ordinary & Partial Differential Equations, Linear Algebra, Communication System, Digital Signal Processing, Digital Electronics, Power Electronics, Control Systems, Microprocessors and Embedded Systems, Analog Integrated Circuits and Design, VLSI Circuits and Design*

Research Interests

• Robotics • Autonomous Systems • Multi-Robot Coordination • Motion Planning • Path Planning • Real-Time Control • Sensor Fusion • Embedded Systems • Robot Perception • Human-Robot Interaction • Field Deployment of Robotic Platforms

Research Contributions

- **Production multi-AGV fleet — centralized job management + decentralized path coordination.** Designed two complementary layers shipping together as the production fleet stack: **(i)** *FleetCore*, a fail-closed ROS 2 coordinator handling FIFO job dispatch from physical call-stations, per-robot sequence-counter FSMs (IDLE / WORKING / RECOVERING / INACTIVE), mutual exclusion on shared unload resources, layered timeouts at each mission gate (accept, arrival, pickup / unload), staggered re-dispatch to recover from concurrent obstacle stalls, and a heartbeat / directory / failsafe substrate giving the operator a single-button fleet-wide override; and **(ii)** a fully decentralized A* path-planning protocol in which each robot publishes its plan as a sequence of timestamped grid waypoints and every peer treats those plans as spatiotemporal reservations subject to three conflict checks (node conflict, edge swap, final-cell occupation), with continuous timestamp maintenance between re-plans and plan-preserving obstacle handling. Validated by months of zero-collision production operation on a 3-robot fleet in a live human-occupied factory. *Manuscript in preparation, first / corresponding author.*
- **Bilateral cross-coupled BLDC speed synchronization on a dual-MCU architecture.** A distributed speed-synchronization scheme where each of two BLDC motors is regulated by its own microcontroller running a PID loop with velocity and acceleration feedforward; a bilateral cross-coupled term, evaluated identically on both nodes from a dedicated inter-MCU speed exchange, drives the inter-motor speed difference toward zero. All gains GA-tuned offline against an identified motor model. Outperforms an independent-PID baseline on both transient and steady-state synchronization metrics. *Manuscript under review, first / corresponding author.*
- **Closed-form analytical IK with FK-anchored goal accumulation for a tracked mobile manipulator.** Tele-op platform combining a 7-DOF arm, a 2-DOF anti-flipper arm (which also contributes to the collision-avoidance envelope alongside the carrier), and a differential-drive base. A from-scratch analytical IK solver replaces MoveIt's KDL — deterministic, branch-stable, constant-time. Smooth turret-scaling preserves continuity through the home-pose singularity; an FK-anchored goal accumulator substantially reduces un-commanded cross-axis motion. MoveIt 2 layers on for motion-planned special commands (return-to-rest, named configurations) and per-tick collision validity across arm, anti-flipper, and carrier. Hardware assembly underway; once field-tested, set for UN peacekeeping deployment.
- **Latency-compensated 5-state EKF for visual-fiducial AGV localization.** Per-robot pose fused from wheel-encoder odometry and AprilTag camera fixes. The state includes velocity-bias terms that absorb systematic encoder error over long corridors; camera updates are applied at their true measurement timestamp via a rewind-and-replay step on a bounded encoder history buffer. Two operating modes (Kalman filter while moving, complementary filter while parked) seamlessly hand off through arrival / departure events.
- **Fault-tolerant dual-network feedback communication architecture** providing redundant pathways for coordinating unmanned multi-node robotic platforms in the field. *Manuscript under review, co-author.*
- **Browser-side Pareto optimizer for heavy-lift multirotor sizing.** Published interactive tool that enumerates feasible series-parallel battery configurations against a motor's measured thrust / current / throttle curve, evaluating hover-throttle reachability, per-motor current limits, voltage compatibility, and cruise margin. Runs entirely client-side; designed as a reproducible companion artifact to the heavy-lift drone project.
- **α-Graphyne ballistic nanoribbon FET.** Undergraduate research with Dr. Mahbub Alam modelling quantum transport in 2D-material field-effect transistors. *Published in Micromachines (MDPI), 2023.*

Publications

- **Md Monirul Islam** et al., "Peer-Aware Distributed Path Planning for a Production-Deployed Multi-AGV Fleet in a Human-Occupied Garment Factory", In Preparation, 2026 (first / corresponding author).
- **Md Monirul Islam** et al., "Experimental Speed Synchronization of a Dual-BLDC Tracked Differential-Drive Vehicle Using a GA-Tuned Bilateral Cross-Coupled Controller on a Dual-MCU Architecture", Under Review, 2026 (first / corresponding author).

- **Md Monirul Islam** et al., "A Dual-Network Feedback Communication System for Multi-Node Unmanned Robotic Control", Under Review, 2026 (co-author).
- Md Shakil Tanvir, Fiaz Ahmed Tonmoy, **Md Monirul Islam**, Mainul Islam, Mahamudul Hasan, Raihan Ur Rashid Razeen, "Optimizing Master-Slave Broadcast Communication in Multi-Node Networks Using RS485 Standard", 2024, 27^{th} International Conference on Computer and Information Technology (ICCIT), IEEE. doi: 10.1109/ICCIT64611.2024.11022430
- H. Khan, **Md Monirul Islam**, R.I. Roya, S.N. Azad, "Performance Analysis of an α-Graphyne Nano-Field Effect Transistor", *Micromachines*, MDPI, 2023. doi: 10.3390/mi14071385

Professional Experience

- **Research and Development Unit, Cybernetics Hi-Tech Solutions (Pvt) Ltd** — Dhaka, Bangladesh
  R&D Engineer and Team Lead *(June 2023 – present)*
  - Lead the R&D engineering team taking robotic platforms from concept through field deployment; own architecture, coding, hardware integration, commissioning, and post-deployment support.
  - **CyberFleet (production AGV fleet).** Lead engineer end-to-end. The platform productised the work and is now operating continuously at The Urmi Group RMG facility, Dhaka.
  - **Jontro Soinik v1 / v2 EOD ROVs.** Owned the electrical / firmware side across both generations. v1 was gifted to the Peruvian Armed Forces by the Bangladesh Army and is deployed under the UN Peacekeeping Mission in Mali (2024); v2 was Army-validated (disrupter and IP55 tests) and is set for Congo (2025). Ran on-site operator-training sessions and field hand-over to receiving units.
  - **Jontro Soinik 2.0 (7-DOF mobile manipulator).** Owned the control / software stack for the tracked tele-op platform.
  - **Heavy-lift multirotor (50 kg payload).** Owned BMS, propulsion sizing, frame, and power-distribution PCBs; shipped the sizing study as an interactive in-browser tool on the portfolio site.
  - **SD15-N target-tracking surveillance drone.** Designed the autonomous-tracking pipeline (vision → MAVLink → Pixhawk).
  - **Titan 550 industrial CoreXY 3D printer.** Two-iteration product build; the in-house fab loop that produces fixtures, brackets, and prototypes for every other platform on this list.
  - **In-house PCB & CNC.** Designed all controller / communication boards across the ROV & AGV platforms in Altium / KiCad and milled them in-house on a 3-axis CNC — no external fab.
  - **Other:** hybrid VTOL quad-plane prototype; real-time pet-food sorting line for an industrial-automation client.
- **Ulkasemi Pvt. Limited** — Dhaka, Bangladesh
  Intern (Dec 2022 – Mar 2023)
  - Worked on IC design using TSMC 180 nm and GPDK 90 nm technologies.
  - Developed a Wishbone interface and validated a high-speed BiFET Op-Amp in Verilog.
- **Walton Hi-Tech Industries Limited** — Dhaka, Bangladesh
  Industrial Trainee (Nov 2022)
  - Observed and analyzed PCB manufacturing and EV assembly workflows.
  - Gained exposure to industrial electronics production workflows.

Selected Undergraduate Projects

- **Real-time traffic speed estimation with YOLOv2**: Trained YOLOv2 in MATLAB to detect vehicles and estimate per-vehicle speed from monocular highway video.
- **6-DOF robotic arm (analytical IK)**: Designed in SOLIDWORKS, 3D-printed entirely on a self-built hobby printer using custom 3D-printed planetary gearboxes; closed-form analytical inverse-kinematics control on stepper motors.
- **Autonomous GPS-waypoint delivery drone**: Mechanical design in SOLIDWORKS, Arduino flight controller with PID stabilisation, GPS-waypoint navigation, and a custom payload-deployment mechanism.
- **Self-built hobby 3D printer**: Designed every component in SOLIDWORKS, fabricated the frame on a laser cutter, Arduino Mega controller running custom Marlin firmware. The workhorse that produced every other undergraduate project's parts.
- **RV32I multicycle RISC-V processor + Wishbone memory controller**: Custom Verilog implementation with full testbench verification; physical-layout exploration in Cadence at TSMC 180 nm.
- **OMR machine from discrete digital logic**: Optical-mark recognition built entirely from discrete NAND / RAM / MUX / Comparator / LDR components — no microcontroller in the loop.
- **Regulated boost converter (TSMC 180 nm, Cadence)**: Schematic capture and physical layout of a voltage-regulated DC-DC boost converter.

- **Closed-loop induction-motor speed control**: V/f scalar control with SVPWM inverter modulation in MATLAB / Simulink.
- **Particle-concentration light-scattering sensor**: Laser + photodetector light-scattering setup with MATLAB-side calibration curves and signal-conditioning circuit.
- **Real-time environmental monitoring**: Solar-powered IoT node logging temperature, humidity, and CO_2 over GSM; Arduino + Autodesk Eagle PCB.
- **Laboratory variable DC power supply**: Linear-regulator design with current limiting and short-circuit protection; MATLAB efficiency study.
- **Electrical-services design (seven-storey building)**: End-to-end electrical design: load estimation, panel scheduling, cable sizing, lighting / power layouts, riser diagram, earthing scheme.

Skills Summary

- **Robotics & Middleware**: ROS 2 (Jazzy), MoveIt 2, ros2_control, JointTrajectoryController, URDF / Xacro / SRDF, AprilTag, MAVLink, dronekit, pymavlink
- **Embedded & MCU**: STM32 (HAL & LL), ESP32, AVR, Arduino, ARM Cortex-M, µVision Keil, BigTreeTech 32-bit boards
- **Control & Algorithms**: Closed-form analytical IK, Extended Kalman Filter, PID (cascaded, GA-tuned), S-curve motion profiling, distributed A* path planning, peer-aware spatiotemporal reservations
- **Flight Control & Vision**: ArduPilot, iNav, Betaflight, Pixhawk, YOLOv11n, OpenCV
- **Simulation & Design Software**: MATLAB / Simulink, Cadence Virtuoso, PSPICE, PSAF, Icarus Verilog, GTKWave
- **PCB & Circuit Design**: Altium Designer, KiCad, Autodesk Eagle, Proteus; in-house 3-axis CNC milling for prototype boards
- **CAD & Mechanical Design**: SolidWorks
- **3D Printing & CNC**: Marlin (customised), Ultimaker Cura, Creality CR-6 SE; FlatCAM, Universal G-code Sender, GRBL
- **Programming Languages**: Python, C, Embedded C, Verilog
- **Documentation**: Microsoft Office Suite, LaTeX, Jekyll (al-folio)
- **Soft Skills**: Leadership, Project Management, Field Commissioning, Operator Training, Technical Writing

Honors and Awards

- University Merit Scholarship, BUET (three times — January 2019, January 2020, January 2022)
  - Awarded to students with outstanding academic record (GPA 3.90+) in a semester
- Dean's List Award, BUET (four times — all academic years)
  - Awarded to students obtaining an average GPA of 3.75+ in two consecutive semesters of an academic year
- University Stipend (three times — January 2018, July 2018, July 2021)
- Higher Secondary Certificate (HSC-2017) Board Scholarship
- Runners-up, Bangabandhu Creative Talent Hunt 2014, Pabna Division

Extra-curricular Activities

- Affiliate Member, BUET Robotics Society (BRS). (Jan 2022 – Mar 2023)
- Affiliate Member, Team Interplanetar – BUET Mars Rover Team. (Jun 2022 – Mar 2023)

Field Deployments

- 3-robot AGV fleet – live production at The Urmi Group RMG facility, Dhaka, 2025 – present
- Jontro Soinik v1 EOD ROV – UN Peacekeeping Mission, Mali, 2024 (Bangladesh Army → Peruvian Armed Forces handover)
- Jontro Soinik v2 EOD ROV – tested by Bangladesh Army, set for UN Peacekeeping deployment in the Republic of Congo, 2025

References

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